Problem Set 3

Gradients and least squares, orthogonality, springs, and the first steps in time

18.085 Computational Science and Engineering I • Fall 2026 • G. G. Rozenman

Due Friday, November 13, 2026, at 11:59 pm ET

Covers Lectures 8–13 (Strang §1.6, 2.1–2.3): positive definite matrices and the Hessian, the heat equation, QR and Gram–Schmidt, least squares, springs and masses, and oscillations with forward Euler. Six problems, 165 points: 25, 25, 20, 25, 30 and 40.

Collaboration. You are encouraged to work together, but all final write-ups must be your own. List your discussion partners and any outside resources you consulted, including AI tools. Using an AI system in place of your own reasoning without saying so violates the collaboration policy; using one as a study aid and saying so does not.

Computing. Problems 2(d) and 6(d) are computational. Use Python or MATLAB, your choice, starting from `pset3_starter.py` or `pset3_starter.m` on Canvas. Attach your code, your tables and your plots to the PDF.

1. Least squares by calculus: gradients and Hessians (25 pts) ★★⋆

Lecture 11 found the best fit from a right angle: the error is perpendicular to the columns. Here you reach the same equations with the tools of Lecture 8, the gradient and the Hessian. Vectors are columns, and ∇f is the column of partial derivatives $\partial f/\partial v_1, \dots, \partial f/\partial v_n$.

(a) A linear function. For a fixed $\boldsymbol{a} \in \mathbb{R}^n$ let $f(\boldsymbol{v}) = \boldsymbol{a}^{\mathsf{T}}\boldsymbol{v}$. Show that $\nabla f = \boldsymbol{a}$. What is the Hessian of f?

(b) A quadratic form. For a symmetric $\boldsymbol{S} \in \mathbb{R}^{n\times n}$ let $f(\boldsymbol{v}) = \boldsymbol{v}^{\mathsf{T}}\boldsymbol{S}\boldsymbol{v} = \sum_{i,j} S_{ij} v_i v_j$. Show that $\nabla f = 2\boldsymbol{S}\boldsymbol{v}$ and that the Hessian is $2\boldsymbol{S}$. Check both for $\boldsymbol{S} = \left[\begin{smallmatrix} 2 & -1 \\ -1 & 3 \end{smallmatrix}\right]$ by writing $f(v_1, v_2)$ out and differentiating. If $\boldsymbol{S}$ is not symmetric, what is $\nabla(\boldsymbol{v}^{\mathsf{T}}\boldsymbol{S}\boldsymbol{v})$?

(c) For $\boldsymbol{A} \in \mathbb{R}^{m\times n}$ and $\boldsymbol{b} \in \mathbb{R}^m$ let $E(\boldsymbol{v}) = \|\boldsymbol{A}\boldsymbol{v} - \boldsymbol{b}\|^2$. Show that

$$E(\boldsymbol{v}) = \boldsymbol{v}^{\mathsf{T}}\boldsymbol{A}^{\mathsf{T}}\boldsymbol{A}\boldsymbol{v} - 2\,\boldsymbol{b}^{\mathsf{T}}\boldsymbol{A}\boldsymbol{v} + \boldsymbol{b}^{\mathsf{T}}\boldsymbol{b},$$

and use (a) and (b) to find ∇E and the Hessian of E.

(d) The minimum. Setting $\nabla E = \boldsymbol{0}$ gives the normal equations $\boldsymbol{A}^{\mathsf{T}}\boldsymbol{A}\,\bar{\boldsymbol{v}} = \boldsymbol{A}^{\mathsf{T}}\boldsymbol{b}$ of Lecture 11. Assume the columns of $\boldsymbol{A}$ are independent. Use the Hessian test of Lecture 8 to explain why $\bar{\boldsymbol{v}}$ is a minimum. Then show directly that $E(\bar{\boldsymbol{v}} + \boldsymbol{d}) = E(\bar{\boldsymbol{v}}) + \|\boldsymbol{A}\boldsymbol{d}\|^2$ for every $\boldsymbol{d}$. Why does this make $\bar{\boldsymbol{v}}$ the *only* minimum, and a global one? Finally, if $\boldsymbol{A} = \boldsymbol{Q}\boldsymbol{R}$ as in Lecture 10 ($\boldsymbol{Q}^{\mathsf{T}}\boldsymbol{Q} = \boldsymbol{I}$, $\boldsymbol{R}$ square, upper triangular and invertible), show that the normal equations reduce to $\boldsymbol{R}\bar{\boldsymbol{v}} = \boldsymbol{Q}^{\mathsf{T}}\boldsymbol{b}$.

(e) Weighted least squares. Three thermometers in the same water bath read 20.4, 19.8 and 20.1 °C. Their stated uncertainties are 0.1, 0.2 and 0.2 °C, so we weight each squared error by $c_i = 1/\sigma_i^2$ and choose the temperature T that minimises

$$E_C(T) = \sum_{i=1}^{3} c_i\,(b_i - T)^2 = (\boldsymbol{A}T - \boldsymbol{b})^{\mathsf{T}}\boldsymbol{C}\,(\boldsymbol{A}T - \boldsymbol{b}), \qquad \boldsymbol{A} = \begin{bmatrix}1\\1\\1\end{bmatrix}, \quad \boldsymbol{C} = \operatorname{diag}(c_1, c_2, c_3).$$

Repeat (c) and (d) for $E_C(\boldsymbol{v}) = (\boldsymbol{A}\boldsymbol{v} - \boldsymbol{b})^{\mathsf{T}}\boldsymbol{C}(\boldsymbol{A}\boldsymbol{v} - \boldsymbol{b})$ with $\boldsymbol{C}$ symmetric positive definite, and show that the minimiser solves $\boldsymbol{A}^{\mathsf{T}}\boldsymbol{C}\boldsymbol{A}\,\bar{\boldsymbol{v}} = \boldsymbol{A}^{\mathsf{T}}\boldsymbol{C}\boldsymbol{b}$. (This is the $\boldsymbol{A}^{\mathsf{T}}\boldsymbol{C}\boldsymbol{A}$ of Lecture 12.) Compute $\bar{T}$ for the thermometers and compare it with the plain average.

2. Gram–Schmidt, QR and least squares (25 pts) ★★⋆

Let

$$\boldsymbol{A} = \begin{bmatrix}1 & 9\\ 4 & 0\\ 8 & 9\end{bmatrix}, \qquad \boldsymbol{b} = \begin{bmatrix}14\\11\\13\end{bmatrix}.$$

(a) Apply Gram–Schmidt to the columns $\boldsymbol{a}_1, \boldsymbol{a}_2$ of $\boldsymbol{A}$ to find orthonormal $\boldsymbol{q}_1, \boldsymbol{q}_2$, and write $\boldsymbol{A} = \boldsymbol{Q}\boldsymbol{R}$ with $\boldsymbol{Q}$ 3×2 and $\boldsymbol{R}$ 2×2 upper triangular. Check that $\boldsymbol{Q}^{\mathsf{T}}\boldsymbol{Q} = \boldsymbol{I}$ and $\boldsymbol{Q}\boldsymbol{R} = \boldsymbol{A}$.

(b) $\boldsymbol{A}\boldsymbol{x} = \boldsymbol{b}$ has no solution (why not?). Use Problem 1(d), or show again, that the normal equations $\boldsymbol{A}^{\mathsf{T}}\boldsymbol{A}\bar{\boldsymbol{x}} = \boldsymbol{A}^{\mathsf{T}}\boldsymbol{b}$ are equivalent to $\boldsymbol{R}\bar{\boldsymbol{x}} = \boldsymbol{Q}^{\mathsf{T}}\boldsymbol{b}$. Find $\bar{\boldsymbol{x}}$ by back substitution, and check your answer in the normal equations themselves. Also check that $\boldsymbol{A}^{\mathsf{T}}\boldsymbol{A} = \boldsymbol{R}^{\mathsf{T}}\boldsymbol{R}$.

(c) Find the projection $\boldsymbol{p} = \boldsymbol{A}\bar{\boldsymbol{x}}$ and the error $\boldsymbol{e} = \boldsymbol{b} - \boldsymbol{p}$. Check that $\boldsymbol{e}$ is perpendicular to both columns of $\boldsymbol{A}$, and find the distance from $\boldsymbol{b}$ to the plane spanned by the columns. Compute the 3×3 matrix $\boldsymbol{P} = \boldsymbol{Q}\boldsymbol{Q}^{\mathsf{T}}$ and verify that $\boldsymbol{P}\boldsymbol{b} = \boldsymbol{p}$ and $\boldsymbol{P}^2 = \boldsymbol{P}$. Lecture 10 showed that $\boldsymbol{Q}\boldsymbol{Q}^{\mathsf{T}} \neq \boldsymbol{I}$ for a tall $\boldsymbol{Q}$; what does $\boldsymbol{I} - \boldsymbol{P}$ do to a vector?

(d) Computational: why software uses QR. Take

$$\boldsymbol{A}_\epsilon = \begin{bmatrix}1 & 1\\ \epsilon & 0\\ 0 & \epsilon\end{bmatrix}, \qquad \boldsymbol{b} = \begin{bmatrix}3\\1\\0\end{bmatrix}.$$

Show, by solving the 2×2 normal equations by hand with ϵ kept as a symbol, that

$$\bar{\boldsymbol{x}} = \frac{1}{\epsilon(2+\epsilon^2)}\begin{bmatrix}1 + 3\epsilon + \epsilon^2\\ 3\epsilon - 1\end{bmatrix}.$$

For $\epsilon = 10^{-1}, 10^{-2}, \ldots, 10^{-10}$, solve the least-squares problem on the computer in two ways: (i) form $\boldsymbol{A}^{\mathsf{T}}\boldsymbol{A}$ and $\boldsymbol{A}^{\mathsf{T}}\boldsymbol{b}$ and solve the normal equations; (ii) compute the reduced QR factorisation (`numpy.linalg.qr(A)` or `qr(A,0)`) and solve $\boldsymbol{R}\boldsymbol{x} = \boldsymbol{Q}^{\mathsf{T}}\boldsymbol{b}$. Make a table of the relative errors $\|\boldsymbol{x} - \bar{\boldsymbol{x}}\| \,/\, \|\bar{\boldsymbol{x}}\|$. Explain what you see: what happens to the entry $1 + \epsilon^2$ of $\boldsymbol{A}^{\mathsf{T}}\boldsymbol{A}$ in double precision ($\epsilon_{\text{mach}} \approx 2.2 \times 10^{-16}$), and how is the condition number of $\boldsymbol{A}^{\mathsf{T}}\boldsymbol{A}$ related to that of $\boldsymbol{A}$? (The condition number is $\sigma_{\max}/\sigma_{\min}$, the ratio of the largest to the smallest singular value; the singular values of $\boldsymbol{A}_\epsilon$ are $\sqrt{2+\epsilon^2}$ and ϵ, the square roots of the eigenvalues of $\boldsymbol{A}_\epsilon^{\mathsf{T}}\boldsymbol{A}_\epsilon$.)

3. Orthogonal polynomials (20 pts) ★★⋆

Polynomials of degree at most 2 form a vector space of dimension 3: every such polynomial is $c_0 + c_1 x + c_2 x^2$, so the monomials $1, x, x^2$ are a basis. They are not the only basis, and with an inner product we can ask for an orthonormal one. Here we use the inner product and norm

$$\langle p, q \rangle = \int_0^1 p(x)\, q(x)\, dx, \qquad \|p\| = \sqrt{\langle p, p \rangle}.$$

(a) Show that $6x^2 - 6x + 1$ is orthogonal to 1 and to x, and hence to every polynomial of degree at most 1. Are 1 and x orthogonal to each other?

(b) Apply Gram–Schmidt to $1, x, x^2$, in that order, to find an orthonormal basis $\ell_0(x), \ell_1(x), \ell_2(x)$.

(c) Let $p(x) = 2x + 1$. Compute $\|p\|$ from the integral, then find the coefficients a_k in $p = a_0\ell_0 + a_1\ell_1 + a_2\ell_2$ by inner products (Lecture 10: coordinates in an orthonormal basis are dot products). Check that $a_0^2 + a_1^2 + a_2^2 = \|p\|^2$.

(d) **The best quadratic.** Find the polynomial q^* of degree at most 2 that minimises $\|x^3 - q\|$. Do it twice:

- project onto your orthonormal basis: $q^* = \sum_k \langle x^3, \ell_k \rangle\, \ell_k$;
- write $q = c_0 + c_1 x + c_2 x^2$ and solve the normal equations $\boldsymbol{H}\boldsymbol{c} = \boldsymbol{d}$, where $H_{ij} = \langle x^i, x^j \rangle$ and $d_i = \langle x^3, x^i \rangle$ for $i, j = 0, 1, 2$.

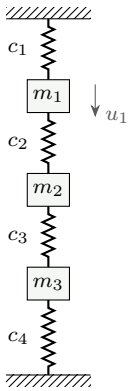
Check that the two answers agree, that the error $x^3 - q^*$ is orthogonal to 1, x and x^2, and find $\|x^3 - q^*\|^2$. The matrix $\boldsymbol{H}$ is the 3×3 *Hilbert matrix*; the $n \times n$ version has condition number about 5×10^2 for $n = 3$, 1.5×10^7 for $n = 6$ and 1.6×10^{13} for $n = 10$. In one or two sentences, relate this to Problem 2(d): which of the two methods would you use to fit a polynomial of degree 9?

4. A hanging chain: $\boldsymbol{A}^{\mathsf{T}}\boldsymbol{C}\boldsymbol{A}$, energy and determinacy (25 pts) ★★⋆

Three masses hang in a line between a ceiling and a floor, joined by four springs with constants

$$\boldsymbol{c} = (c_1, c_2, c_3, c_4) = (2, 1, 1, 2) \text{ N/mm},$$

spring 1 at the ceiling and spring 4 at the floor. Each mass weighs 4 N, so $\boldsymbol{f} = (4, 4, 4)$ N. As in Lecture 12, $\boldsymbol{u}$ is the downward displacement of the masses (in mm), $\boldsymbol{e} = \boldsymbol{A}\boldsymbol{u}$ the elongations, $\boldsymbol{w} = \boldsymbol{C}\boldsymbol{e}$ the spring forces (positive for tension), and force balance is $\boldsymbol{A}^{\mathsf{T}}\boldsymbol{w} = \boldsymbol{f}$.



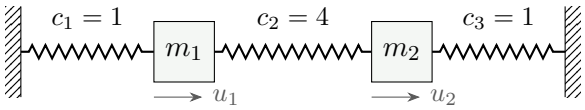
(a) Write down $\boldsymbol{A}$ (4×3) and $\boldsymbol{C}$, and compute the stiffness matrix $\boldsymbol{K} = \boldsymbol{A}^{\mathsf{T}}\boldsymbol{C}\boldsymbol{A}$. Show that $\boldsymbol{u}^{\mathsf{T}}\boldsymbol{K}\boldsymbol{u} = \sum_j c_j e_j^2$ and conclude that $\boldsymbol{K}$ is positive definite.

(b) Solve $\boldsymbol{K}\boldsymbol{u} = \boldsymbol{f}$. Find $\boldsymbol{e}$ and $\boldsymbol{w}$ and check that $\boldsymbol{A}^{\mathsf{T}}\boldsymbol{w} = \boldsymbol{f}$. Which springs are stretched and which are compressed? With what force does the chain pull on the ceiling, and with what force does it push on the floor? Check that the two add up to the total weight.

(c) Minimum energy. The potential energy is $P(\boldsymbol{u}) = \frac{1}{2}\boldsymbol{u}^{\mathsf{T}}\boldsymbol{K}\boldsymbol{u} - \boldsymbol{f}^{\mathsf{T}}\boldsymbol{u}$. Use Problem 1(a,b) to find its gradient and Hessian, and explain why the equilibrium of (b) is the unique minimum of P. Show that $P(\boldsymbol{u}^*) = -\frac{1}{2}\boldsymbol{f}^{\mathsf{T}}\boldsymbol{u}^*$ at the equilibrium $\boldsymbol{u}^*$ and evaluate it (N mm = mJ). Then evaluate P at the trial displacement $(4, 4, 4)$ mm and check that the difference equals $\frac{1}{2}\boldsymbol{d}^{\mathsf{T}}\boldsymbol{K}\boldsymbol{d}$ for the right $\boldsymbol{d}$.

(d) Cut the floor spring. Now remove spring 4, so the bottom mass hangs free. $\boldsymbol{A}$ is 3×3. Show that the spring forces $\boldsymbol{w}$ follow from $\boldsymbol{A}^{\mathsf{T}}\boldsymbol{w} = \boldsymbol{f}$ alone, before you know any c_j. Then find $\boldsymbol{e}$ and $\boldsymbol{u}$. Why could you not do this in (b)? (Lecture 12: statically determinate and indeterminate.)

5. Two masses in motion: normal modes (30 pts) ★★⋆



Two masses $m_1 = m_2 = 1$ kg slide without friction between two walls. The outer springs have constant 1 N/m and the middle spring 4 N/m. u_1, u_2 are the displacements to the right from equilibrium. With masses in kg and spring constants in N/m, time is in seconds.

(a) Write Newton's law for each mass separately (Lecture 13) and show that $\ddot{\boldsymbol{u}} + \boldsymbol{K}\boldsymbol{u} = \boldsymbol{0}$ with $\boldsymbol{K} = \begin{bmatrix} 5 & -4 \\ -4 & 5 \end{bmatrix}$. Check that $\boldsymbol{K} = \boldsymbol{A}^{\mathsf{T}}\boldsymbol{C}\boldsymbol{A}$ for the fixed–fixed $\boldsymbol{A}$ of Lecture 12.

(b) Find the eigenvalues and orthonormal eigenvectors of $\boldsymbol{K}$, and the frequencies of the two normal modes. Sketch or describe each mode. Explain from the springs why the middle spring does not affect the lower eigenvalue at all, and write the higher eigenvalue in terms of c_1 and c_2.

(c) At $t = 0$ the masses are released from rest at $\boldsymbol{u}(0) = (2, 0)$ cm. Write $\boldsymbol{u}(0)$ in the eigenvector basis and find $\boldsymbol{u}(t)$. When does mass 1 first pass through its equilibrium position? Where are the masses at $t = \pi$? Is the motion periodic, and with what period?

(d) Energy. Let $E(t) = \frac{1}{2}\dot{\boldsymbol{u}}^{\mathsf{T}}\dot{\boldsymbol{u}} + \frac{1}{2}\boldsymbol{u}^{\mathsf{T}}\boldsymbol{K}\boldsymbol{u}$. Show that $dE/dt = 0$; where did you use $\boldsymbol{K} = \boldsymbol{K}^{\mathsf{T}}$? Compute E for the motion in (c) (in mJ), and show that it splits into one part per mode. What fraction of the energy is in the fast mode?

(e) A first-order system. With $\boldsymbol{v} = \dot{\boldsymbol{u}}$, write the motion as $\boldsymbol{y}' = \boldsymbol{G}\boldsymbol{y}$ for $\boldsymbol{y} = (\boldsymbol{u}, \boldsymbol{v}) \in \mathbb{R}^4$, with a 4×4 block matrix $\boldsymbol{G}$. Show that if $\boldsymbol{K}\boldsymbol{q} = \omega^2\boldsymbol{q}$, then $(\boldsymbol{q},\ i\omega\boldsymbol{q})$ and $(\boldsymbol{q},\ -i\omega\boldsymbol{q})$ are eigenvectors of $\boldsymbol{G}$ with eigenvalues $\pm i\omega$. List the four eigenvalues of $\boldsymbol{G}$. What do purely imaginary eigenvalues say about the size of $e^{\boldsymbol{G}t}\boldsymbol{y}(0)$ (Lecture 7)?

6. Forward Euler: stable and unstable time steps (40 pts) ★★★

Forward Euler (Lecture 13) replaces $\boldsymbol{y}' = \boldsymbol{G}\boldsymbol{y}$ by

$$\frac{\boldsymbol{y}_{n+1} - \boldsymbol{y}_n}{\Delta t} = \boldsymbol{G}\boldsymbol{y}_n, \qquad \text{that is} \qquad \boldsymbol{y}_{n+1} = (\boldsymbol{I} + \Delta t\, \boldsymbol{G})\, \boldsymbol{y}_n.$$

Lecture 13 called the time step h; here h is the grid spacing in space, so the time step is Δt. Only forward Euler is needed in this problem.

(a) Decay. For $y' = -ay$ with $a > 0$ (for instance an RC circuit with $a = 1/RC$), show that $y_n = (1 - a\Delta t)^n y_0$. For which values of $a\Delta t$ does the numerical solution (i) decay without changing sign, (ii) decay while changing sign at every step, (iii) grow? Take $a = 20\ \mathrm{s}^{-1}$ and $y_0 = 1$, and compute the numerical value at $t = 0.6\,\mathrm{s}$ for $\Delta t = 0.04$, 0.075 and $0.12\,\mathrm{s}$ (15, 8 and 5 steps). Compare with the exact e^{-12}.

(b) The heat equation. The rod of Lecture 9, $u_t = \gamma u_{xx}$ on $0 < x < 1$ with $u = 0$ at both ends, becomes on the grid $x_j = jh$, $h = 1/(n+1)$,

$$\frac{d\boldsymbol{u}}{dt} = -\frac{\gamma}{h^2}\boldsymbol{K}_n\boldsymbol{u}.$$

Take $n = 49$ (so $h = 0.02$) and $\gamma = 0.01$. The eigenvalues of $\boldsymbol{K}_n$ are $\mu_k = 4\sin^2(k\pi/(2n+2))$, $k = 1, \ldots, n$, with discrete sine eigenvectors (Lecture 9). Explain why forward Euler on this system is forward Euler on n separate equations of the form in (a), one per eigenvector, with $a_k = \gamma\mu_k/h^2$. (Hint: $\boldsymbol{K}_n = \boldsymbol{V}\boldsymbol{\Lambda}\boldsymbol{V}^{\mathsf{T}}$.) Find the largest stable Δt and compare it with $h^2/(2\gamma)$. Which eigenvector sets the limit, and what does it look like on the grid? The slowest mode decays like $e^{-a_1 t}$: about how long until it has dropped to 1%, and how many steps of the largest stable size does that take? If h is halved, by what factor does the total work grow?

(c) Oscillation. Apply forward Euler to the system $\boldsymbol{y}' = \boldsymbol{G}\boldsymbol{y}$ of Problem 5(e). Write $\boldsymbol{u}_n = \alpha_1\boldsymbol{q}_1 + \alpha_2\boldsymbol{q}_2$ and $\boldsymbol{v}_n = \beta_1\boldsymbol{q}_1 + \beta_2\boldsymbol{q}_2$ in the eigenvectors of $\boldsymbol{K}$. Show that each mode steps separately,

$$\alpha_{n+1} = \alpha_n + \Delta t\, \beta_n, \qquad \beta_{n+1} = \beta_n - \Delta t\, \omega^2 \alpha_n,$$

and that its energy $\frac{1}{2}(\beta^2 + \omega^2\alpha^2)$ is multiplied by exactly $1 + \omega^2\Delta t^2$ at every step. Conclude that forward Euler is unstable for an undamped oscillator for *every* $\Delta t > 0$. For the motion of Problem 5(c) with $\Delta t = \pi/300$ (600 steps per period 2π), predict $E(2\pi)/E(0)$. Which mode is responsible for most of the growth? Roughly how small must Δt be to keep the energy error below 1% over one period, and how many steps is that?

(d) Computational.

(i) Heat equation of (b) with $u_j(0) = 1$ for $j = 16, \ldots, 29$ (the nodes strictly between 0.3 and 0.6) and $u_j(0) = 0$ otherwise. Run forward Euler to $t = 2.1$ with $\Delta t = 0.0175$ (120 steps) and with $\Delta t = 0.021$ (100 steps). Plot both results together with the exact solution $e^{-(\gamma/h^2)\boldsymbol{K}_n t}\boldsymbol{u}(0)$, computed from the eigendecomposition as in Problem Set 2. Explain the shape of the result for $\Delta t = 0.021$ using (a) and (b): which eigenvectors grow, and by about what factor over the 100 steps does the fastest one grow?

(ii) Two masses of Problem 5, starting from $\boldsymbol{u}(0) = (2, 0)$, $\boldsymbol{v}(0) = \boldsymbol{0}$. Run 600 forward Euler steps with $\Delta t = \pi/300$. Plot u_1 against the exact $\cos t + \cos 3t$, and plot E_n/E_0 against t. Compare the final value with your prediction in (c).